ECO111: Quiz 4

April, 2024
Total Points: 25, Time: 40 minutes.

Note: While answering the questions below, keep in mind that it is not enough to just write down the final answers. You need to provide arguments supporting the answers, and only then will any credit be awarded.

Question 1

You have been hired as a portfolio manager for a stock brokerage. Your first job is to invest 1 unit of money in a portfolio of two assets. The first asset is a safe asset with a sure return of 8% interest. The second asset is a risky asset with a 25% expected rate of return, but the standard deviation of this return is 10%. Your client wants a portfolio with as high a rate of return as possible consistent with a standard deviation no larger than 4%. How much of her money do you invest in the safe asset? [**2 points**]

Question 2

Anthony, who is a merchant in Vizag, owns just one ship. The ship is worth 200 million rupees. If the ship sinks, Anthony loses 200 million rupees. The probability that it will sink is 0.02 . Anthony's total wealth including the value of the ship is 225 million rupees. He is an expected utility maximizer with von Neuman-Morgenstern utility $U(W)$ equal to the square root of W. What is the maximum amount that Anthony would be willing to pay in order to be fully insured against the risk of losing his ship? [**3 points**]

Question 3

Suppose that a honey farm is located next to an apple orchard and each acts as a competitive (that is, as a price taking) firm. Let the amount of apples produced be measured by A and the amount of honey produced be measured by H. The cost function of the honey firm is $c_H(H) = H^2/100$ and the cost function of the apple firm is $c_A(A) = A^2/100 - H$. The price per unit of honey is 2 and the price per unit of apples is 3.

1. If each firms operate independently, what is the equilibrium amount of honey produced? What is the equilibrium amount of apples produced? [**2 points**]

2. Suppose the honey and apple firms merged. What would be the profit-maximizing output of honey for the combined firm? What would be the profit-maximizing amount of apples? [**2 points**]

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Question 4

A firm is contemplating to hire a salesman who would be entrusted with the task of selling an air fryer. The hired salesman is efficient with probability 0.25 and inefficient with probability 0.75 and there is no way to tell, by looking at the salesman, if he is efficient or not. An efficient salesman can sell the air fryer with probability 0.8 and an inefficient salesman can sell the air fryer with probability 0.4. The firm gets revenue equal to Rs. 1000 if the air fryer is sold and gets nothing if it is not sold. In either case, however, the salesman has to be paid a wage of Rs. 100.

1. Calculate the expected profit of the firm if the salesman is hired. [**2 points**]

2. Suppose that instead of the fixed payment of Rs.100, the firm pays a commission of $t\%$ on its sales to the salesman (i.e., if the good is sold the salesman gets Rs. $1000 \times \frac{t}{100}$ and nothing if the good remains unsold). A salesman, irrespective of whether he is efficient or inefficient, has an alternative option of working for Rs. 80. A salesman knows whether he is efficient or not and cares only about the expected value of his income. Find the value of t that will maximize the expected profit of the firm. [**6 points**]

Question 5

Suppose Anjali has to decide how much time z to devote to a risky business venture. With probability π, the business is a success and generates a monetary income for Anjali of $(1+r)z$, where $r > 1$. With probability $1 - \pi$, the business is failure, in which case Anjali's monetary income is $\frac{z}{2}$. Notice that the more time Anjali devotes to the business venture, the more she makes. Anjali can choose any amount of time z between zero and an upper limit L.

1. Suppose Anjali chooses z to maximize the difference between the expected value of her income and the amount of time z , as long as this difference is positive, and otherwise chooses zero time. For what values of r would Anjali choose zero time? If r is larger than the value(s) that cause Anjali to choose $z = 0$, what value of z will Anjali choose? [**4 points**]

2. Now suppose that instead Anjali maximizes the difference between the expected utility of her income and the amount of time z. The utility she derives from income x is given by $\ln(x)$. For what values of r does Anjali choose $z = 0$? For larger values of r, what value of z does Anjali choose? [**4 points**]

ECO111: Quiz 3

March, 2024
Total Points: 25, Time: 40 minutes.

Note: While answering the questions below, keep in mind that it is not enough to just write down the final answers. You need to provide arguments supporting the answers, and only then will any credit be awarded.

Question 1

If a country's income has been growing steadily at 2% per year and now stands at \$50,000, what would have been its income 200 years ago? [**3 points**]

Question 2

If the interest rate is $r = 10\%$ and will remain $r = 10\%$ forever, then a bond that will pay 95 rupees a year forever, starting one year from now, is worth how many rupees today? [**3 points**]

Question 3

Sehwag's paintings are unappreciated now. Nobody is willing to pay anything to have them on the walls. In 5 years Sehwag's works will become popular. People will suddenly be willing to pay, beginning from year 5 onwards, 1,000 rupees a year to have original paintings by Sehwag on their walls and will continue to be willing to do so ever after. If investors realize that this is the case and if the interest rate is and always will be r, a painting by Sehwag will currently be worth how much? [**3 points**]

Question 4

For every two parathas that she consumes, Shalu insists on having one bowl of curd. She does not however insist on consuming the same amount every week. Her utility function is

$$U = \min\{p_1, 2c_1\} \min\{p_2, 2c_2\}$$

where p_1 and p_2 are the number of parathas she consumes this week and next week, and c_1 and c_2 are the number of bowls of curd she consumes this week and next. Parathas cost 2 rupees per piece, and curd costs 1 rupee per bowl. Shalu has a present value of 100 rupees to spend on these goods in the two weeks. The weekly interest rate is 1%. How many parathas will Shalu will consume this week? [**6 points**]

Question 5

In an isolated desert village, the only crop is bajra. This year (year 1) the harvest will be 1,000 units. Next year (year 2) it will be 150 units. There is no bajra left over from previous years. There is no trade with the outside world. Bajra can be stored from one year to the next, but rats will eat 25% of what is stored in the village in a year. The villagers have preference given by $U(c_1, c_2) = c_1 c_2$ where c_1 is consumption of bajra this year, and c_2 is consumption of bajra next year.

1. How much bajra will the villagers consume this year? How much bajra will the villagers consume next year? [**4 points**]

2. Suppose now that a road is built to the village so that now the village is able to trade with the rest of the world. Now the villagers are able to buy and sell bajra at the world price, which is 1 per unit for each of the two years. They are also able to borrow and lend money at an interest rate of 10%. Now, how much bajra will the villagers consume this year? How much bajra will the villagers consume next year? [**4 points**]

3. Suppose now that all is as in part 2. of the question except that there is also in addition a per unit transportation cost of 0.10 for every unit of bajra hauled into or out of the village. Under this scenario, draw a "budget line" showing consumption possibilities for the village, with this year's consumption on the horizontal axis and next year's consumption on the vertical axis. Make sure to accurately label the intercepts of the "budget line". [**2 points**]

$c_1 = 1000 - s \cdot \frac{25}{100}$

$c_2 = 150 + \frac{75}{100}(s)$

$p_1 c_1 + \frac{p_1 c_2}{1+r} = 100$

$c_1 + \frac{c_2}{1+r} = 1000$

$\frac{25}{100}$ $c_1 = (1000 - s) + b$

$c_2 = 150 + \frac{75}{100}(s) = b(1+r)$

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ECO111: Midterm

February, 2024
Total Points: 25, Time: 120 minutes.

Note: While answering the questions below, keep in mind that it is not enough to just write down the final answers. You need to provide arguments supporting the answers, and only then will any credit be awarded.

Question 1

In Borduria, a committee of three voters, 1, 2, and 3, must decide how to share a resource of total value 1. One of the voters makes a proposal[1]. The proposal is then voted upon. If a majority, including the proposer, approves it, the proposal is implemented. If not, a voter[2] makes a proposal and the three vote again using the same procedure. If the second proposal is rejected, a voter[3] proposes and then the three vote again. Suppose that the committee has a maximum limit of three proposals. If the third proposal is not accepted, the resource is lost and everyone receives 0. In addition, the resource shrinks with each failed proposal (depreciates over time). The first proposal thus concerns the division of a resource worth 1, the second of a resource worth $d < 1$, and by the third proposal the resource is only worth $d \times d = d^2$. Assume that if indifferent over accepting or rejecting a proposal, a voter will accept. Suppose also that instead of making a proposal in order of their index, each has an equal probability of 1/3 of being the next proposer. Note that this means that even if a voter's proposal gets rejected in a particular round, he has a chance of getting to make the proposal in the next round (if a next round exists). In equilibrium, how much of the resource does the voter who makes the first proposal get? [**5 points**]

Question 2

Sumeria is a country with two citizens, Akhenaten (citizen A) and Nefertiti (citizen N). The government is considering building a pyramid that will cost 1000. Let B_A and B_N be the true benefits to A and N respectively of having the pyramid built. Unfortunately the government does not know the actual value of the true benefits, only that each is non-negative and is less than 1000. The government devices a policy with the following features to decide whether the pyramid should be built or not:

1. The government asks A and N to report their benefits of the pyramid being built: denote the values reported by them as R_A and R_N respectively. Each of the reported benefits have to be non-negative and less than 1000.

2. The pyramid will be built if and only if $R_A + R_N$ is greater than or equal to the cost of building the pyramid.

3. If the pyramid is build, the government will impose a tax of $1000 - R_A$ on N and a tax of $1000 - R_N$ on A.

[1] a proposal specifies how the current value of the resource is to be divided among the three voters - the shares need not be integers.

[2] who may or may not be the same person as the one who made the initial proposal

[3] who may or may not be the same person as the one who made the initial or the person who made the second proposal

The government knows that each citizen will seek to maximise their net benefit, that is the difference between their true values and the tax which must be paid if and only if the pyramid is build. It further knows that if a citizen is indifferent between telling the truth and lying, she/he will tell the truth. If the pyramid is not build, A and N each have a payoff of 0. The Sumerian government's belief is that under this policy, each of the citizens will truthfully report their true benefits. Is the Sumerian government's belief correct? [**NOTE**: It is not enough to enough guess one or other. You need to provide a proof of the statement if you think the government's belief is correct, or a counter-example if you think it is incorrect, and only then will any credit be awarded.] [**5 points**]

Question 3

Consider a seller who is interested in selling a single private value good. There are two potential buyers, each with a valuation that is drawn independently and uniformly from the interval [0, 1]. The seller will offer the good using a second-price sealed-bid auction, but she can set a "reserve price" of $r \geq 0$ that modifies the rules of the auction as follows. If both bids are below r then neither bidder obtains the good and it is destroyed. If both bids are at or above r then the regular auction rules prevail. If only one bid is at or above r then that bidder obtains the good and pays r to the seller. What is the optimal reserve price for the seller if she wishes to maximize her expected profit? [**5 points**]

Question 4

You are interested in buying a company called Ajeeb Paradox. You know that their marketing division is inefficient, and hence if you take over the company, you can increase its value by 75% of whatever it is worth before. However, it's accounting division is excellent, so that the company's true value is hard for outsiders to identify. Insiders know the true value perfectly. You suspect that the company's value in the hands of its current management is somewhere between 10 crore and 110 crore, uniformly distributed over this range. The current management will sell the company to you, if and only if, your bid exceeds the true value known to them. If your offer is rejected, you don't get the company and don't have to pay anything. What should you bid if you want to maximize your expected profit? [**5 points**]

Question 5

Manoj, Navjot, and Ravindra are having a shootout. On each round, until only one player remains alive, the current shooter can choose one of the other players as target and is allowed one shot. At the start of the game, they draw straws to see who goes first, second, and third, and they take turns repeatedly in that order. A player who is hit is killed. Manoj is a perfect shot, Navjot has 80% accuracy, and Ravindra has 50% accuracy. We assume that players are not required to aim at an opponent and can simply shoot in the air on their turn[4], if they so desire. What is the probability that Manoj will survive the shootout? What is the probability that Navjot will survive the shootout? What is the probability that Ravindra will survive the shootout? [**5 points**]

[4] no one is hit by such a shot

ECO111: Quiz 2

February, 2024
Total Points: 25, Time: 30 minutes.

Note: While answering the questions below, keep in mind that it is not enough to just write down the final answers. You need to provide arguments supporting the answers, and only then will any credit be awarded.

Question 1

A college is trying to fill **one remaining seat** in its undergraduate program in applied probability. It judges the merit of any applicant by giving her an entrance test. It is known that there are **three** interested applicants who will apply **sequentially**. The college has to make decision for an applicant immediately after observing the result of the applicant's test, and without seeing the results of future applicants. If the college admits the first applicant, it cannot admit the second or the third applicant. If it rejects the first applicant, it must admit the second or the third. It is not possible to delay a decision on the first applicant till the second or third applicant is tested. Similarly, if the college rejects the first applicant, and admits the second applicant, it cannot admit the third. If it rejects the first and second applicant, it must admit the third. It is not possible to delay a decision on the second applicant till the third applicant is tested. Before an applicant is tested, it is known that her likely mark is an independent continuous random variable drawn from the uniform distribution between 0 and 100. If the college wants to maximize the expected mark of its admitted student, what is the lowest mark for which the college should admit the first student? [**5 points**]

Question 2

A monopolist's cost of producing q units of output is given by $C(q) = 2q$. The monopolist can potentially sell her product in two distinct markets. In market 1, demand as a function of unit price is $D_1(p) = 9 - p$. In market 2, demand as a function of unit price is $D_2(p) = 4 - p$. The monopolist can choose to sell in none, one or both markets. If it chooses to sell in both markets, it **must** set the same price in both markets. What should the monopolist do to maximize profit? [Note: Be specific in your answer - mention the markets where the monopolist should sell, and also what should be the optimal prices and quantities it chooses in those markets.] [**5 points**]

Question 3

Consumers differ in their willingness to pay for the device, WID. No one needs more than one. All consumers value owning a WID more highly, the larger is the total number of consumers using such devices. Denote the expected total number of WID users by n^e, which we can also call the "expected size" of the WID network. If all consumers expect the size of the WID network to be n^e, and the price of the device is p, then the number of users who will want to buy the device (i.e. the total quantity demanded) is given by $n = 80 - p + \frac{1}{5}n^e$, where. Suppose WIDINC is the only seller of WID and that per unit cost per WID is 20. Suppose also that consumers are quite sophisticated and form accurate expectations about the size of the WID network, for any price p that might be set, so that n must equal n^e. What is the maximum profit that WIDINC can earn? **[5 points]**

Question 4

You see an ad for a second hand car you are interested in. The owner has not set a price, but invites potential buyers to make offers. Your prepurchase inspection gives you a rough idea of the value of the car - you think it is equally likely to be anywhere in the range between 1 lakh to 5 lakh rupees[1]. The current owner knows the exact value and will accept your offer only if it exceeds that value. If your offer is rejected, you don't get the car and don't have to pay anything. If your offer is accepted and you get the car, then you will find out the truth. Fortunately, having spent time at a top technology institute, you have some special repair skills and know that when you own the car, you will be able to improve it and increase its value by one-third (that is by 33.3...%) of whatever it is worth. What is the highest offer you can make so that your expected payoff on the deal is non-negative? **[5 points]**

Question 5

The demand curve facing a monopolist is $D(p) = 300/p$ if p is 100 or smaller and $D(p) = 0$ for $p > 100$. The monopolist has a constant cost of 5 per unit produced and no other costs. What is the profit-maximising quantity for this monopolist? **[5 points]**

[1] Thus your calculation of the average value is 3 lakh rupees.

ECO 111: Quiz 1

January, 2024

Total Time: 30 minutes
Total Points: 25 points

Note: While answering the questions below, keep in mind that it is not enough to just write down the final answers. You need to provide arguments supporting the answers, and only then will any credit be awarded.

Question 1

Three pirates must divide 100 gold coins. The coins cannot be cut into pieces[1]. Pirate A is the strongest, followed by Pirate B, followed by Pirate C. Because of ancient pirate tradition, the coins are divided in the following manner. First Pirate A proposes a division of the coins. The three pirates then vote on whether to accept the proposed division. If the proposal gets a majority[2] vote, it is accepted, and the game is over. If the proposal fails to get a majority vote, Pirate A is executed. It is then Pirate B's turn to propose a division of the coins between the two remaining pirates. The same rules apply, with one exception: if the vote is a tie (which can happen this time, because the number of pirates is now even), Pirate B, being the strongest remaining pirate, gets an additional vote to break the tie. Use backward induction to figure out what Pirate A should propose. While answering this question, assume that if someone is indifferent between accepting or rejecting a proposal, then that person will vote in favour of the proposal. [**5 points**]

Question 2

Ten countries are considering fighting global warming. Each country must choose to spend an amount x_i to reduce its carbon emissions, where $0 \leq x_i \leq 1$. The total benefits produced by these expenditures equal twice the total expenditures: $2(x_1 + \cdots + x_{10})$. Each country receives $\frac{1}{10}$ of the total benefits. The ith country's payoff function is its benefits minus its expenditures. Find the pure strategy Nash Equilibrium outcome. [**5 points**]

[1] This means that no one can be offered a non-integer amount of coins.
[2] that is, if more than half the votes are in favour of the proposal

Question 3

Consider a seller who is interested in selling a single private value good. There are two potential buyers, each with a valuation that can take on one of three values, $\theta_i \in \{0, 1, 2\}$, each value occuring with a probability of 1/3. The players' values are independently drawn. The seller will offer the good using a second-price sealed-bid auction, but she can set a reserve price of $r \geq 0$ that modifies the rules of the auction as follows. If both bids are below r then neither bidder obtains the good and it is destroyed. If both bids are at or above r then the regular auction rules prevail. If only one bid is at or above r then that bidder obtains the good and pays r to the seller.

1. What is the expected revenue of the seller if she sets $r = 0$? **[2.5 points]**
2. What is the expected revenue of the seller if she sets $r = 1$? **[2.5 points]**

Question 4

Player 1 is a constable who must decide whether to patrol the market or to relax in the police station. His payoff from relaxing in the police station is 10, while his payoff from patrolling the market depends on whether he catches a pickpocket, who is player 2. If the pickpocket prowls the market, then the constable (if he is patrolling the market) will catch him and obtain a payoff of 30. If the pickpocket stays in his home, then the constable's payoff from patrolling the market is 0. The pickpocket must choose between staying home or prowling the market. If he stays at home then his payoff is 0, while if he prowls the market his payoff is -15 if the constable is patrolling the market and 10 if the constable is relaxing in the police station. Find all (pure strategy and mixed strategy) Nash equilibria of this game. **[5 points]**

Question 5

Ashwin, Bishan, and Chetan are having a shootout. On each round, until only one player remains alive, the current shooter can choose one of the other players as target and is allowed one shot. At the start of the game, they draw straws to see who goes first, second, and third, and they take turns repeatedly in that same order. A player who is hit is killed. Ashwin is a perfect shot[3], Bishan has 80% accuracy, and Chetan has 50% accuracy.

Suppose, that as a result of draws the order is Bishan gets to shoot first, followed by Ashwin, followed by Chetan. What is the probability that Ashwin will survive the shootout? What is the probability that Bishan will survive the shootout? What is the probability that Chetan will survive the shootout? **[5 points]**

[3]which means he never misses